

How realistic is synthetic oncology EHR data for real-world evidence?

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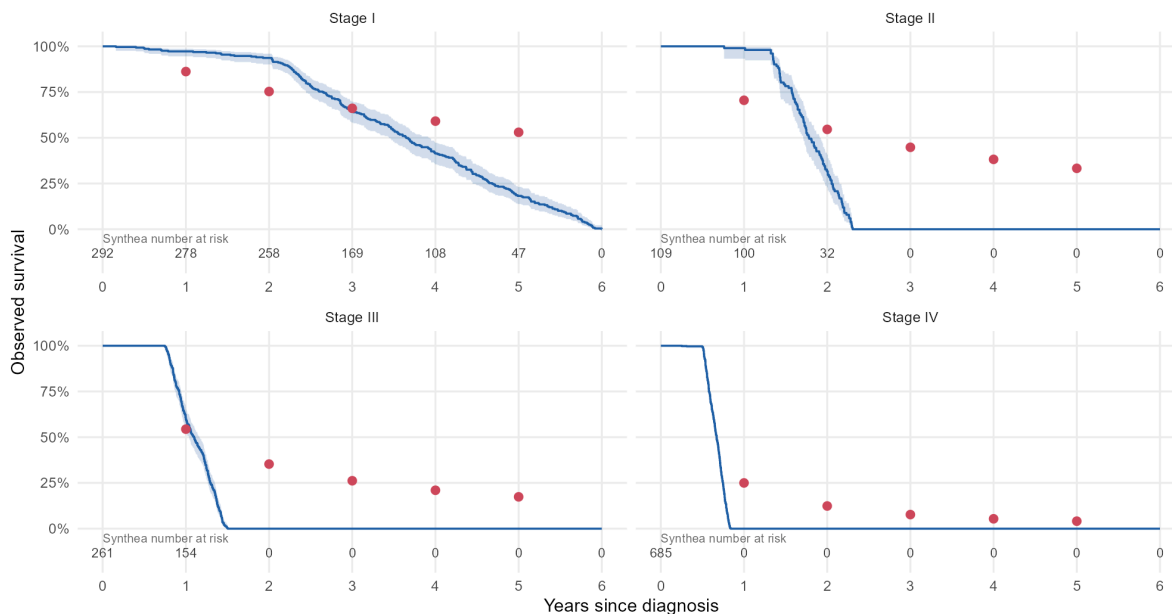
Question. Does synthetic EHR data (Synthea) reproduce real-world non-small cell lung cancer (NSCLC) survival well enough for RWE method development, judged against SEER registry aggregates?

Data and pipeline. Synthea (synthetic EHR) → OMOP CDM 5.4 (ETL-Synthea) → PostgreSQL → dbt → R survival analysis → SEER 17 (Nov 2025) benchmark; 1,347 staged NSCLC patients aged 50 and over.

Three findings

1. **Summary numbers can look right while the curves are wrong.** The stage III median is 13.2 vs 14 months in SEER, and stage I 5-year RMST is 42.9 vs 43.5 months. But stage I survival at 5 years is 18.2% in Synthea against 53.0% in SEER, and stages II–IV fall to 0% by design (scripted death windows).
2. **Stage ordering is right but the effect is exaggerated.** Cox hazard ratios against stage I are 22.2, 290 and 9,490 (near-complete separation); proportional hazards are rejected.
3. **No age or sex signal, and no treatment contrast.** Age HR 1.05 per 10 years and male HR 1.01; reweighting to the SEER age/sex mix leaves stages II–IV unchanged. 1,346 of 1,347 patients received the same first line, so treatment effects are not identifiable.

Lesson: validate the whole distribution, not summary statistics.



SEER 17, observed survival — Synthea (Kaplan-Meier, 95% log-log CI)

Synthea: staged NSCLC, age ≥ 50 at diagnosis, $n = 1347$. SEER 17: NSCLC diagnosed 2010-2015, age 50+; error bars are SEER's exported 95% CI.

Synthetic EHR plus public registry aggregates only; no real patient data. Code and results: <https://github.com/erickyegon/oncology-rwe-nsclc>. Medium article: <https://medium.com/@keyegon/from-seer-stat-to-r-a-practical-guide-to-extracting-and-validating-cancer-survival-data-2601dd584b88>.